TAMENESS OF A FINITE SEPARABLE EXTENSION OF A DISCRETE VALUATION RING PASSES TO INTERMEDIATE SUBEXTENSIONS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted proof span contains the author's sentence 'Since kappa(m')/kappa_A is finite separable, the same is true for kappa(m)/kappa_A.' and its short closing line, both delivered verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Multiplicativity of ramification indices in a tower of extensions of discrete valuation rings. invoked at proof lines 34176-34177. Transfers the prime-to-residue-characteristic condition from M down to L.
- The discussion of finite separable extensions of discrete valuation rings, including the behaviour of integral closures. referred to at proof lines 34168-34171. Supplies the framework of maximal ideals and residue field extensions used throughout.
- Surjectivity of the spectrum map of a finite ring extension. used at proof lines 34172-34175. Produces a maximal ideal of the integral closure lying over a given maximal ideal.

Source reference. Original lemma 32952: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32952–32952. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33006: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33006–33006. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K. If $M/L/K$ are finite separable extensions and M is tamely ramified with respect to A, then L is tamely ramified with respect to A.*

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Proof. We will use the results of the discussion in Remark 33006 without further mention. Let $C/B/A$ be the integral closures of A in $M/L/K$. Since C is a finite ring extension of B, we see that $\mathrm{Spec}(C) \to \mathrm{Spec}(B)$ is surjective. Hence for ever maximal ideal $\mathfrak{m} \subset B$ there is a maximal ideal $\mathfrak{m}' \subset C$ lying over $\mathfrak{m}$. By the multiplicativity of ramification indices (Lemma 32952) and the assumption, we conclude that the ramification index of $B_{\mathfrak{m}}$ over A is prime to the residue characteristic. Since $\kappa(\mathfrak{m}')/\kappa_A$ is finite separable, the same is true for $\kappa(\mathfrak{m})/\kappa_A$. $\square$